

Shawn Liu

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TECHNICAL SKILLS

Languages: C, C++, C#, Python, JavaScript, SQL, Java, TypeScript, HTML & CSS, React, R, Bash
Developer Tools: Git, Linux, Jira, Microsoft SQL Server, Valgrind, DB2, Cypress, Playwright, Azure
Design Models: ASP.NET, MVC, OOP, UML
Libraries: Pandas, NumPy, Jupyter, MatLab, scikit-learn, PyTorch, OpenGL

WORK EXPERIENCES

Fortinet Oct 2025 — Present
Full Stack Web Developer Burnaby, BC

- Architected and deployed **autonomous AI agents** with custom tool-calling capabilities, enabling dynamic task execution and external API integrations for the company's flagship AI feature.
- Refactored the AI software architecture of the company by migrating from high-parameter models (GPT-4.1) to lightweight and efficient alternatives (Qwen3.5, Gemma-4), reducing the inference latency by **27%** and lowering computational costs while maintaining comparable accuracy for core AI feature functionality.
- Resolved 30+ bugs in company software.

Bosda Inc. Sept. 2024 – Dec. 2024
Software Engineering Intern Aurora, ON

- Constructed 40+ interfaces in **C# ASP.Net** for company databases, allowing company data to be manipulated through interfaces instead of direct access to the database.
- Applied **Sequential, LSTM, Dense** and **TimeseriesGenerator** models for predicting the sales and price of company product, increasing the purchase rate of company products **40%**.
- Implemented robust **APIs** to integrate machine learning functionalities in Python into company codebase, which allowed company's codebase to develop enhanced data processing and automation capabilities.

SparkLease Inc. Sept 2023 – Dec. 2023
Full-Stack Developer North York, ON

- Designed and implemented recommendation algorithm for product listing, improving user click-through rate of products by **10%**.
- Constructed 4 Websites using **ASP.Net** with **MVC** structure to display various user data for admin purposes.
- Created WebJobs in **C#** on the **Microsoft Azure Portal** that automates the programs to calculate and classify user data for developing recommendation algorithm.
- Designed and organized database in **Microsoft SQL Server** to keep track of user information and statistics for application functionalities.

EDUCATION

University of Waterloo Sep. 2020 – May 2025
Bachelor of Honours Computer Science Waterloo, ON

- **Minors:** Combinatorics and Optimization + Computational Mathematics.
- **Relevant Courses:** OOP, Data Structures, Algorithms, Operating Systems, User Interfaces, HCI, Machine Learning, Artificial Intelligence, Databases, Neural Networks, Computer Vision, Computer Graphics.

PROJECTS

Exam Generator | Node.js, React, TypeScript, Python

- Web application that generates custom University of Waterloo Math and Computer Science exams using Qwen Model API. Built with React frontend and Python backend
- Implemented PDF export functionality for generated exams, enabling easy printing and distribution.

AutoSched | React, Python, JavaScript, CSS

- Calendar application that takes in a natural language input of a event and schedules all tasks associated with the event.